

Adult Age Differences in On-line Sentence Processing of Korean Relative Clauses^{*}

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Kim, Meesook and Noh, Soo Rim. 2019. Adult age differences in on-line sentence processing of Korean relative clauses. *Discourse and Cognition* 26:3, 23-42. The goal of this study was to investigate age-related differences in processing relative clause sentences, using a self-paced reading paradigm. In this study, we used four types of Korean relative clauses (RCs): two types of subject RCs (SS and OS) and two types of object RCs (SO and OO). The results showed that compared with younger adults, older adults in general had much longer reading times, longer response times, and low accuracy in processing all types of relative clause sentences. Second, the results further showed that older adults had more difficulty in processing the OS-type and the OO-type RCs, which involved a garden-path effect, than the SS-type and SO-type RCs. These findings are consistent with the claim that older adults have greater difficulty processing syntactically complicated or ambiguous sentences. Therefore, we concluded that older adults' greater difficulty in processing RCs was related with reduced cognitive resources. (Sangji University and Chungnam National University) (146 words)

Key words: aging, relative clause, garden-path effect, working memory, sentence comprehension

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1. Introduction

There is considerable evidence that the ability to comprehend sentences becomes less accurate and slower among healthy older adults (Waters and Caplan 2005; Stine-Morrow, Noh, and Shake 2010; Caplan, DeDe, Waters, and Michaud 2011). Aging has generally been found to result in the decline in the cognitive capacities such as working memory or general speed of cognitive processing. Previous studies of language comprehension by older adults have shown that participants found it particularly difficult to comprehend and process syntactically complex sentences such as relative clauses (RCs) because these structures have higher demands for working memory resources (Kemper and Herman 2006). Their difficulties can be explained by the decline in their ability to temporarily store or retain in working memory some linguistic elements during sentence processing (King and Just 1991). The findings suggest that the decline in their working memory capacity is a major reason for their difficulty in sentence comprehension (DeDe, Caplan, Kemtes, and Waters 2004; Waters and Caplan 2005; Caplan et al. 2011; DeDe 2015). Nevertheless, there has not been much research on the relationship in Korean. Thus, the current study examines how sentence comprehension of relative clause sentences is affected by older and younger adults in Korean.

RCs have been one of the most widely studied syntactic structures in the psycholinguistic literature (Frauenfelder, Segui, and Mehler 1980; Frazier 1987; Kim 1995; Mecklinger, Schriefers, Steinhauer, and Friederici 1995; Schriefers, Friederici, and Kuhn 1995; Cohen and Mehler 1996; Kanno and Nakamura 2001; Miyamoto and Nakamura 2003; Kwon, Polinsky, and Kluender 2004; Kwon, Kluender, Kutas, and Polinsky 2013). There are different types of RCs, depending on the function of the head noun phrase (NP). First, the subject relative clause in English when the head noun, *the reporter*, is the subject of the verb in the relative clause, as shown in (1a). Second, the object relative clause in English when the head noun, *the reporter*, is the object of the verb in the relative clause, as shown in (1b).

(1) Subject RCs in English

a. The reporter that attacked the senator admitted the error.

Object RCs in English

b. The reporter that the senator attacked admitted the error.

It has been generally accepted that subject RCs are easier to process than object RCs in English (King and Just 1991; Gibson 1998; Gordon, Hendrick, and Johnson 2001; Traxler, Morris, and Seely 2002). The difference between these clause types can be described in several ways. For example, the subject relative clause sentence in (1a) forces few processing demands on the readers because the subject of the main clause, *the reporter*, is also the subject of the embedded relative clause. In contrast, the object relative clause sentence in (1b) requires the reader to assign the correct syntactic relations because the subject of the main clause, *the reporter*, must also be interpreted as the object of the embedded clause (Traxler et al. 2002). Another simple explanation for this difficulty is that the greater number of intervening words between the head noun and the gap in object RCs make them more difficult to process than subject RCs (Gibson 1998; Grodner and Gibson 2005). King and Just (1991) reported that processing of object RCs required greater working memory than processing of subject RCs.

Meanwhile, the typological literature has addressed the issue of whether the choice of relative clause placement (head-initial vs. head-final) affects the processing efficiency. In head initial relative clause in English and related languages (typically those in Subject-Verb-Object word order or SVO), the head noun, *kica* (reporter), always precedes the gap, as in (1). In contrast, in head final relative clause in Korean and related languages (typically those in Subject-Object-Verb word order or SOV), the head noun, *kica* (reporter), always follows its gap as shown in (2)¹⁾.

(2) Subject RC in Korean

- a. [__ uywon-ul kongkyekha-n] kica-ka silswu-lul
 senator-ACC attack-ADN reporter-NOM error-ACC
 siinhayssta
 admitted
 ‘The reporter who attacked the senator admitted the error.’

b. Object RC in Korean

- [uywon-i __ kongkyekha-n] kica-ka silswu-lul siinhayssta
 senator-NOM attack-ADN reporter-NOM error-ACC admitted
 ‘The reporter who the senator attacked admitted the error.’

1) We use the following abbreviations in glossing our examples: ACC: accusative, ADN: adnominal, DECL: declarative, NOM: Nominative, PRES: present tense, PST: past tense.

It has been an issue of whether the cross-linguistic difference with respect to the relative clause ordering of the head noun and its gap affects the processing of the relative clause sentences. Many cross-linguistic studies have confirmed the claim that regardless of the different ordering of the head noun and its gap, subject-gap RCs are easier to process than object-gap RCs, which is commonly called “the subject RC preference”, by using a variety of psycholinguistic measures (Frauenfelder et al. 1980; Frazier 1987; Kim 1995; Mecklinger et al. 1995; Schriefers et al. 1995; Cohen and Mehler 1996; Kanno and Nakamura 2001; Miyamoto and Nakamura 2003; Kwon et al. 2004; Kwon et al. 2013). For instance, Kwon et al. (2013) conducted an event-related brain potential (ERP) study of Korean RCs to test whether the subject preference could be found in SOV languages like Japanese and Korean. They suggested that brain responses to RCs were very similar in SVO and SOV languages, by supporting that there was a universal subject-gap preference in the processing of RCs.²⁾

Previous studies have further claimed the effect of individual working memory capacity in processing RCs (King and Just 1991; Just and Carpenter 1992). Just and Carpenter (1992) have suggested that the working memory resources possessed by individuals were limited and thus, syntactic processing would deteriorate if working memory load were increased by such factors as syntactic complexity. One instance of such effects was that younger adults with high working memory spans processed syntactically complex sentences like object relative clauses more efficiently than individuals with low working memory spans. Thus, they interpreted this as evidence for the effect of working memory capacity limits in language processing.

Most previous studies on relative clause constructions have so far focused on sentence comprehension and working memory with younger adults exclusively (King and Just 1991; McDonald, Pearlmutter, and Seidenberg 1994; Caplan and Waters 1999; Waters and Caplan 2005). In contrast, relatively few studies have examined comprehension of RCs in aging (King and Just 1991; Kemper, Crow, and Kemptes 2004; Waters and Caplan 2005; Kemper and Herman 2006; Kemper and Liu 2007; Caplan et al. 2011; DeDe 2015). In a self-paced reading study, Caplan et al. (2011) reported that both younger and older adults indicated longer processing times for object RCs than for subject RCs, but this effect of syntactic complexity was greater in older adults than that was in

2) For more detailed cross-linguistic patterns in RCs, please refer to Kwon et al. (2013).

younger adults. In another self-paced listening study, Waters and Caplan (2005) also found that older adults were slower and less accurate than younger adults. Furthermore, the eye-movement study by Kemper and Liu (2007) indicated that older and younger adults with low working memories showed longer fixation times to process object relative clause sentences. In addition, older adults showed more difficulty than young adults while reading relative clause sentences. Thus, Kemper and Liu (2007) concluded that age-associated declines in working memory affected syntactic processing.

We have so far seen that there were a lot of experimental studies on the processing of RCs with younger adults across languages. Meanwhile, research on the relationship between aging and working memory has been conducted exclusively with English older adults. Few studies have so far examined the age differences in on-line sentence processing of Korean RCs with older adults. Therefore, the current study is the first to investigate how older adults and younger adults comprehend all types of Korean RCs, which involve syntactic complexity and ambiguities, using a self-paced reading paradigm.

2. Background Literature

RCs have received considerable attention from the areas of syntax, language acquisition and psycholinguistics in Korean (Clancy, Lee, and Zoh 1986; Kim 1995; Kwon et al. 2004; Kwon et al. 2013; Lee 2007). RCs in Korean are typically divided into four different structural types, depending on the grammatical roles of head noun and the gap, as in (3). The current study looked at those in the canonical SOV order to test the four types of relative clause sentences, as shown in (3a-d):

- (3) a. SS-type (Subject head noun; Subject gap)
 [uywon-ul kongkyekha-n] enlonin-i phyencipcang-ul
 [senator-ACC attack-ADN] journalist-NOM editor-ACC
 silheha-n-ta.
 dislike-PRES-DECL
 ‘The journalist who attacked the senator disliked the editor.’
 b. SO-type (Subject head noun; Object gap)
 [uywon-i kongkyekha-n] enlonin-i phyencipcang-ul
 [senator-NOM attack-ADN] journalist-NOM editor-ACC

silheha-n-ta

dislike-PRES-DECL

‘The journalist who the senator attacked disliked the editor.’

c. OS-type (Object head noun; Subject gap)

enlonin-i [uywon-ul kongkyekha-n] phyencipcang-ul

journalist-NOM [senator-ACC attack-ADN] editor-ACC

silheha-n-ta

dislike-PRES-DECL

‘The journalist disliked the editor who attacked the senator.’

d. OO-type (Object head noun; Object gap)

enlonin-i [uywon-i kongkyekha-n] phyencipcang-ul

journalist-NOM [senator-NOM attack-ADN] editor-ACC

silheha-n-ta

dislike-PRES-DECL

‘The journalist disliked the editor who the senator attacked.’

It has been pointed out that the OS-type and the OO-type RCs, which have the object head nouns as in (3c) and (3d), led to a garden-path effect in that they could be temporarily ambiguous or confusing by their case markers (Clancy et al. 1986; Kim 1995; Kwon et al. 2004; Kwon et al. 2013; Lee 2007). For example, the OO-type RC in (3d) contained the double nominative construction which could make readers’ processing harder, in spite of the fact that it is acceptable in Korean grammar. In addition, the OS-type RC in (3c) also could make readers’ processing more difficult because the first NP is nominative-marked and the second NP is accusative-marked just as in sentences following the basic word order.³⁾ Thus, the OS-type and the OO-type RCs in the canonical SOV sentence could add the processing cost due to the garden-path effect. In order to avoid such for the processing difficulties caused, previous experimental studies used the so-called ‘scrambled sentences’ in which a direct object was fronted to the beginning of the sentence, as shown in (4a-b) (Clancy et al. 1986; Kim 1995; Kwon et al. 2004; Kwon et al. 2013). That is, instead of the canonical SOV word order sentences for the object head noun types, the scrambled OSV word order sentences for the object head noun types

3) For detailed discussion of scrambled OO and OS RCs, please refer to Kim (1995) and Kwon et al. (2004).

were used as in (4a-b):

- (4) a. Scrambled OSV: OS-type (Object head noun; Subject gap)
[uywon-ul kongkyekha-n] phyencipcang-ul enlonin-i
[senator-ACC attack-ADN] editor-ACC journalist-NOM
silheha-n-ta
dislike-PRES-DECL
‘The journalist disliked the editor who attacked the senator.’
- b. Scrambled OSV: OO-type (Object head noun; Object gap)
[uywon-i kongkyekha-n] phyencipcang-ul enlonin-i
[senator-NOM attack-ADN] editor-ACC journalist-NOM
silheha-n-ta
dislike-PRES-DECL
‘The journalist disliked the editor who the senator attacked.’

In contrast with the previous studies, the current study used the OO-type and the OS-type RCs in the canonical word order to investigate the garden-path effect that could bring the heavy processing loads to younger and older adults (Caplan and Waters 1999; Frazier 1987). Furthermore, it has been suggested that older adults had greater difficulty in processing syntactically ambiguous garden-path sentence than younger adults (Kemper, Crow, and Kemptes 2004; Yoo and Dickey 2019). Thus, we further investigate age-related differences in on-line sentence processing of the OS-type and the OO-type RCs among younger and older adults.

In addition to the garden-path effect, one of the well-known processing approaches for relative clause study is Sheldon’s (1974) “parallel function hypothesis”. Sheldon claimed that people could easily process the sentences in which the grammatical roles of the gap and the head matched. As such, the hypothesis predicted better performance on the SS-type and the OO-type RCs over the SO-type or the OS-type RCs across languages. Sheldon’s (1974) study from English-speaking children indicated that they had no difficulty in understanding the SS-type RCs, whereas they had great difficulty in understanding the SO-type RCs. In addition, Clancy et al.’s (1986) study found that Korean-speaking children more easily processed the SS-type and the OO-type RCs than the SO-type or the OS-type RCs.

Another approach to the RCs has been explained by Keenan and Comrie’s

(1977) NP accessibility hierarchy (subject > object > indirect object > object of a preposition > possessive > object of a comparative), which indicates that grammatical role higher on the accessibility hierarchy is easier to process. Their hierarchy predicts greater ease of processing subject RCs over that of processing object RCs. Thus, the NP accessibility hierarchy claimed that better performance on subject RCs over object RCs across languages could be directly related to the “psychological ease of comprehension”, even though it did not provide a specific account of the cognitive basis. A recent self-paced reading task by Kwon et al. (2004) showed that subject RCs in Korean were easier to process than object RCs, by supporting the NP accessibility hierarchy.⁴⁾

In summary, the parallel function hypothesis by Sheldon (1974) predicts Korean adults’ (both younger and older) better performance on the SS-type and the OO-type RCs than that on the SO-type and the OS-type RCs. Meanwhile, the NP accessibility hierarchy hypothesis by Keenan and Comrie (1977) predicts Korean adults’ (both younger and older) better performance on the SS-type and the OS-type RCs than that on the SO-type and the OO-type RCs. In addition, the garden-path assumption predicts that Korean adults will show more difficulty in processing the unscrambled OS-type and OO-type RCs than the SS-type and the SO-type RCs.

3. Experiment

In this experiment, we examine age-related difficulties in processing RCs and age-related differences among the four types of RCs. The predictions are as follows:

Predictions:

(1) Overall, older adults will show more difficulty in processing RCs than younger adults. For instance, older adults will show longer reaction time, longer reading time, and lower accuracy in processing RCs than younger adults.

(2) The parallel function hypothesis predicts that the SS-type and the OO-type RCs will be easier to process than the SO-type and the OS-type RCs. Meanwhile, the NP accessibility hierarchy predicts that the SS-type and the

4) The NP accessibility hierarchy account based on the psychological basis has been further extended to O’Grady’s (1997) notion of “structural distance” between the gap and the head noun. For more detailed account, please refer to Kwon et al. (2013).

OS-type RCs will be easier to process than the SO-type and the OO-type RCs.

(3) If both younger and older adults show garden-path effects during comprehension of the OO-type and the OS-type RCs, they will have more difficulty in processing such RCs than the SS-type and the SO-type RCs. Furthermore, the garden-path effect on older adults will be significantly different from that on younger adults.

3.1. Participants

40 younger adults (ages 19-25) and 44 older adults (ages 61-81) participated in this experiment. However, one younger adult and 19 older adults were excluded in our analysis due to their extremely low accuracy on the sentence comprehension test (lower than 60%). Thus, the data for 39 younger adults and 25 older adults were used in our study. All participants were right-handed with no neurological disorders. Table 1 summarized the background information of the participants. The education level of older adults showed the distribution of 28% of college graduates or higher, 44% high school graduates and 24% middle school graduates or below. We conducted the Korean Version Mini Mental State Examination (K-MMSE; Kang, Na, and Hahn 1997), which is a brief instrument to assess cognitive abilities. All participants scored at least 25 (cutoff score) out of 30 points on the K-MMSE, and younger adults scored higher than older adults did, $t(62)=4.57$, $p<.001$. Working memory capacity was assessed using a modified version of Lee's (1995) reading sentence span task, which measures concurrent linguistic processing and storage (see Procedures for details). As expected, younger adults had a greater reading span as compared to older adults, $t(62)=6.59$, $p<.001$.

<Table 1> Descriptive Information for Younger and Older adults

	Younger Group M(SD) (N=39)	Older Group M(SD) (N=25)	Group Diff
Gender (% female)	48.7%	44.0%	
Age (years)	21.59 (19.25)	69.32 (61.81)	
Education (% above high school)	100%	75%	**
K-MMSE	28.97 (1.20)	27.48 (1.39)	***
Reading Span	4.16 (0.95)	2.91 (0.55)	***

** $p<.01$, *** $p<.001$

The participants received a monetary stipend for their participation. The study was approved by the Institutional Review Board of C University, South Korea.

3.2. Materials

Test materials were 120 sentences, adapted from Lee (2007). Familiar animals over human names or human professions for their nouns were used because it could be more difficult for older adults to remember three human names or human professions contained in each RC sentence. A sample of the relative clause sentences from our experiment is given in Table 2. The experimental sentences included 24 sentences of each of the four relative clause types: subject-subject (SS), subject-object (SO), object-subject (OS), and object-object (OO). Examples of each of the sentence type are shown in Table 2.

<Table 2> A Sample of Test Sentences

Type of RC	Head Noun	Gap	Test Sentences
SS	Subject	Subject	[oli-lul nemettuli-n] ttokki-ka talamchwi-lul cchochaka-ss-ta. ‘The rabbit that knocked down the duck chased the squirrel’.
SO	Subject	Object	[ttokki-ka nemettuli-n] oli-ka talamcwi-lul ccochaka-ss-ta. ‘The duck that the rabbit knocked down chased the squirrel’.
OS	Object	Subject	talamcwi-ka [ttokki-lul nemettuli-n] oli-lul ccochaka-ss-ta. ‘The squirrel chased the duck [that knocked down the rabbit]’.
OO	Object	Object	talamcwi-ka [oli-ka nemettuli-n] ttokki-lul ccochaka-ss-ta. ‘The squirrel chased the rabbit [that the duck knocked down]’.

24 additional conjunction sentences served as filler sentences (e.g., ‘The giraffe scratched the tiger and he hit the tiger’). The sentences were divided into three blocks such that each block contained 40 sentences, including 32

relative clause sentences (eight for each type) and eight conjunction sentences. Each participant saw three blocks in a randomized order with a short break after each block. For each block of the sentences, two pseudorandomized lists were created with the constraint that no more than two sentences of the same type could appear sequentially. List order within each block was counterbalanced across participants within age groups.

The participants were given a comprehension question to confirm whether the contents of subject case and object case in the main clause, subject case, and object case in the relative clause were correct. For each relative clause type, eight questions relating to these four formats were included. In the case of conjunction sentences, the questions confirming the correctness of the subject or object cases were included in the same portions.

3.3. Procedure

Participants were tested individually in a laboratory setting. After informed consent was obtained, they completed demographic questionnaire and the K-MMSE. The self-paced reading paradigm (E-prime 2.0) was used in this experiment. The reading session was preceded by eight practice sentences and questions. Participants were instructed that they would read a series of short sentences on the computer, one word at a time. Participants were told to read naturally at their own pace, but to read carefully so that they would be able to answer to follow-up questions. Each sentence was presented with a “READY?” signal, and the participant was asked to press the space bar to replace the signal with a plus sign (+) at the top of the left corner of the computer screen where the first word of the sentence would appear. Along with the plus sign, a series of dashes corresponding to the length of each word in the sentence and the appropriate punctuation were presented. When participants pressed the space bar again, the plus sign disappeared, and the first word of the sentence appeared where its respective dashes had been, and so on. Participants were not able to go back to previous words. After each sentence, participants responded to a comprehension question with two choices by pressing the A key with their left index finger if the answer was the first choice and the key with the right index finger if the answer was the second choice. Three blocks of sentences were presented separately with a short break following each block. The participants’ reading time, response time, and accuracy of each comprehension

question were recorded.

After completing the reading session, participants performed the reading sentence span task. For this task, they were presented with a set of sentences (ranging from two to six sentences in each set) and were required to read each sentence aloud while remembering the final word of the sentence. At the end of the set, they were prompted to recall the last word of each sentence. Each set contained two trials, and if the participant recalled the final words correctly at the first trial, s/he was moved to the next level containing one more sentence. If s/he made errors on the second trial of any set, the task was stopped. The span score of the participant was equal to the last set (level) that was successfully recalled plus the value equal to the proportion of all the last words correctly recalled at the next level (Stine and Hindman 1994). For example, given a case where the participant passed the four-sentence set, recalled only three words in the first trial of the five-sentence set and concluded the task by recalling only 4 words in their second try, the reading span was 4.7, calculated by adding the 4 correct sentences to 0.7, as the participants have recalled 7 words out of 10 sentences in the 5-sentence set. The experiment lasted approximately 1.5 hours.

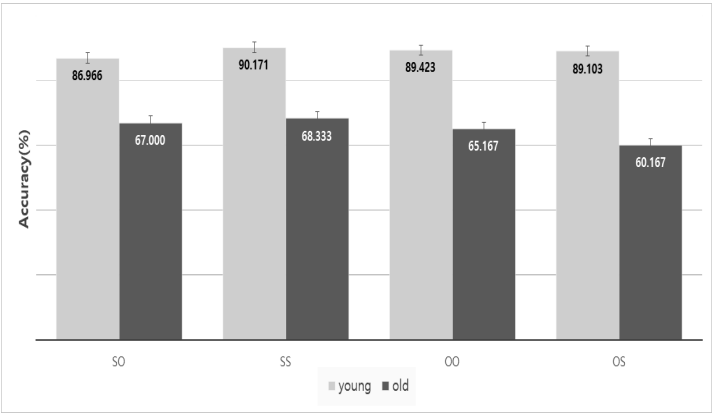
3.4. Results

All analyses of variance (ANOVAs) involving within-subject factors were adjusted using the Greenhouse-Geisser correction for nonsphericity. Whenever appropriate, post hoc comparisons were Bonferroni corrected.

3.4.1. Accuracy

The percentage of correct answerers on the comprehension questions was analyzed using a 2 (Group: younger vs. older) by 4 (Sentence Type: SS, SO, OS, OO) mixed-model ANOVA. Group was a between-subjects factor and Sentence Type was a within-subjects factor. A significant main effect of Group indicated that older adults ($M=65.17$, $SE=1.44$) made more comprehension errors than younger adults did ($M=88.92$, $SE=1.15$), $F(1, 62)=165.36$, $p<.001$, partial $\eta^2=.73$. The effect of Sentence Type was significant, $F(3, 186)=3.35$, $p<.05$, partial $\eta^2=.05$, which was qualified by a significant Group by Sentence Type interaction, $F(3, 186)=3.52$, $p<.05$, partial $\eta^2=.05$. Follow-up between-group comparisons reveal that older adults made more errors than

younger adults for all sentence type conditions ($p<.001$). Within-group comparisons between the sentence type revealed different patterns of comprehension accuracy for the younger and older groups. The interaction is illustrated in Figure 1.



<Figure 1> Relative Clause Type x Age Group

For younger adults, there were no differences in comprehension accuracy across sentence type conditions, Willks’ Lambda=.94, $F(3, 60)=1.31$, $p=.28$, partial $\eta^2=.06$. However, for older adults comprehension accuracy was lowest for the OS-type and it was significantly lower than for the SO-type ($p=.05$) and SS-type ($p<.01$), Willks’ Lambda=.82, $F(3, 60)=4.52$, $p<.01$, partial $\eta^2=.18$.

3.4.2. Response Times

For the analysis of the response times (RTs), RTs greater and less than 3 standard deviations (SDs) from the mean were excluded. The means and standard errors by each group are presented in Table 3. Approximately 1.4% of the younger adults’ data and 0.4% of the older adults’ data were removed. The RT data was analyzed in a 2 (Group) by 4 (Sentence Type) mixed-model ANOVA.

<Table 3> Means (msec) and Standard Errors of Each Group

Sentence Type	Younger	Older	Total
SS	2,798 (265)	6,375 (331)	4,586 (212)

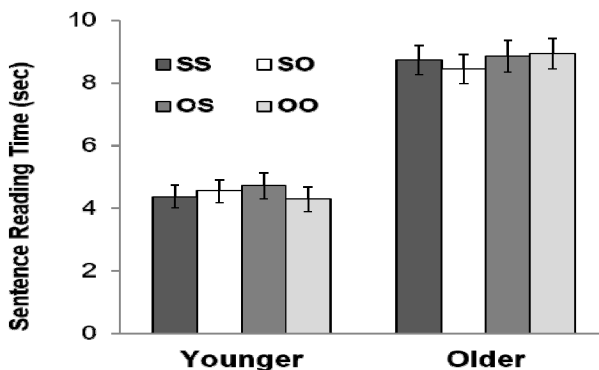
SO	2,838 (266)	6,439 (332)	4,639 (213)
OS	3,229 (300)	7,338 (375)	5,284 (240)
OO	3,267 (291)	6876 (364)	5,072 (233)
Total	3,032 (257)	6,757 (321)	

Overall, older adults were much slower at responding than younger adults and their RTs were more than twice those of the younger adults, $F(1, 62)=82.17$, $p<.001$, partial $\eta^2=.57$. The main effects of Sentence Type was significant, $F(2.66, 164.80)=10.27$, $p<.001$, partial $\eta^2=.14$. The RTs were the longest for the OS-type and shortest for the SS-type. Post-hoc comparisons revealed that RTs in the OO-type and OS-type were significantly longer than in the SO-type and SS-type ($p<.05$). The RTs between the OO-type and OS-type and the RTs between the SO-type and SS-type were not significantly different ($ps=1.00$). The Group by Sentence Type interaction was not significant, $F(2.66, 164.80)=1.48$, $p=.22$, partial $\eta^2=.02$.

3.4.3. Reading Times

Reading time data that were greater and less than 3 SDs from the mean for each sentence type per each participant were excluded from the analysis as outliers. These procedures removed less than 1% of the data for each age group. Younger and older adults' mean sentence reading times for each of the 4 sentence types were analyzed in a 2 (Group) by 4 (Sentence Type) mixed-model ANOVA. Results are illustrated in Figure 2. The main effect of Group was significant, $F(1, 62)=48.61$, $p<.001$, $\eta^2=.44$, indicating that older adults' sentence reading times ($M=8,741$, $SE=477$) were double those of younger adults ($M=4,479$, $SE=382$). The main effect of Sentence Type, $F(2.40, 148.69)=3.22$, $p<.05$, $\eta^2=.03$, was significant, and this effect was further moderated by a significant Group by Sentence Type interaction, $F(2.40, 148.69)=5.13$, $p<.01$, $\eta^2=.08$. Follow-up between-group comparisons show that older adults had longer reading times for sentences than younger adults regardless of the sentence types, $ps<.001$. For younger adults, within-group comparisons between the sentence type revealed a significant difference between the OS-type and the OO-type ($p<.01$), and between the OS-type and the SO-type ($p<.05$), Willks' Lambda=.83, $F(3, 60)=4.03$, $p<.01$, partial $\eta^2=.25$ (see Figure 2). For older adults, there were no significant differences between sentence types for all comparisons, Willks' Lambda=.89, $F(3, 60)=2.42$, $p=.08$,

partial $\eta^2=.11$.



<Figure 2> Online Sentence Reading Time for Each Sentence Type in Younger and Older Groups

4. Discussion and General Conclusions

The primary goal of the current study was to investigate age-related differences in on-line sentence processing of Korean relative clauses. The results for age-related differences showed that there were age-effects in the reading times, response times, and sentence comprehension accuracy in all types of relative clause sentences. Compared with younger adults, older adults exhibited extremely low accuracy (around 60%) and long RTs in comprehension questions. Moreover, older adults took longer to read the relative clause sentences than their younger counterparts. These findings were consistent with the claim that older adults in general had greater difficulty processing syntactically complicated or ambiguous sentence comprehension (Kemtes and Kemper 1997; Waters and Caplan 2005; DeDe 2015). It has been suggested that low accuracy and slowing of online processing times found in older adults' RC sentence comprehension were related to their decline in cognitive resources (Kemper and Liu 2007). Indeed, compared to younger adults, our older adults had a lower reading span, indicating that they have low working memory capacity.

With respect to the 4-types of RCs, an interesting finding of the study was that older adults showed more difficulty in processing the OS-type and the

OO-type RCs than the SO-type and the SS-type RCs.⁵⁾ Older adults' comprehension accuracy in the OS-type RC was significantly lower than it was in the SO-type and the SS-type RCs. In addition, older adults spent more RTs and sentence reading times in the OS-type and the OO-type RCs than in the SS-type and the SO-type RCs. In contrast, younger adults exhibited no significant differences in comprehension accuracy and sentence reading times across sentence types, whereas their RTs in the OS-type and the OO-type were significantly longer than in the SO-type and the SS-type. Accordingly, these results indicate that both younger and older adults showed extreme difficulty in processing the OO-type and the OS-type RCs, compared with the SS-type and the SO-type RCs, even though the processing difficulty for these types in older adults is much greater than that in younger adults. Accordingly, these findings support the assumption that there was a garden-path effect in the unscrambled OS-type or OO-type RC in Korean (Kim 1995; Kwon et al. 2004). In particular, it has been suggested that older adults showed much greater effect on the garden-path effect during the on-line sentence processing than younger adults (Kemper et al. 2004; Yoo and Dickey 2019). Therefore, the findings from this study are consistent with the suggestion that older adults' garden-path effect was greater than younger adults' in on-line measures due to the heavy working memory loads (Yoo and Dickey 2019).

In addition to the garden-path assumption, the NP accessibility hierarchy predicted that subject-gap RCs such as the SS-type and the OS-type RCs, would be easier to process than object-gap RCs such as the SO-type and the OO-type RCs in Korean. The results of this study found that the RT of the SS-type in both younger and older adults was shortest among 4-types, by indicating that the universal subject-gap preference was found in both age groups. In contrast with the prediction, the OS-type was one of the hardest conditions for younger and older adults. This finding could be explained by the garden-path effect in the OS-type. In addition, the parallel function hypothesis predicted that participants would show better performance in processing the SS-type and the OO-type RCs than the SO-type and the OS-type RCs.

5) In contrast, Lee's (2007) study, which used the similar test sentences as we used, found that there was a significant advantage for the OS condition among 4-types of RCs. He mentioned that the OS advantage was an unexpected result under the assumption that there would be a garden-path effect in the OS condition. Nevertheless, he concluded that the OS advantage could be explained by the universal subject-gap preference in RCs.

However, no parallel function effect of head nouns was found in this study in that the RT of the SS-type was not significantly different from the RT of the SO-type.

In sum, the current study examined how older and younger adults comprehend RCs which involved syntactic complexity as well as syntactic ambiguities. Such sentences have been extensively studied in the psycholinguistic literature with younger adults but have received little attention with older adults in Korean. The findings here suggest that aging could be a factor when older adults interpret and process syntactically complicated RCs. Therefore, the results indicate that older adults' greater sentence processing difficulty in RCs may be related to age-related cognitive decline.

Several cautions should be noted regarding the current experiment. First, the number of participants, in particular older adult participants was too lower than intended because a large number of individuals were excluded from the analysis due to their extremely low accuracy on the sentence comprehension test. Second, the experimental sentences used in this study seem to be too difficult for older adults. We used animal names in each experimental sentence, by assuming that they would be perceptually easier to remember than human names and human professions. In contrast with our prediction, using animal kinds in the experimental sentences seems to incur older adults' sentence processing difficulty. Lee and Kwon (2012) suggest that the properties of the noun phrases in RCs, such as the combination of human names-common nouns and the word order of these noun phrases, affect the sentence comprehension. Accordingly, it will be interesting whether older adults will be more sensitive to the properties of noun phrases in Korean RCs than younger adults in further studies. Third, for a more detailed online processing study, eye tracking or ERPs will be required to investigate how aging will affect the ability of processing complex syntactic structures.

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